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### 5.2.2 Distinctive behavior of lithium

The distinctive behaviour of $Li^+$ ion is due to its exceptionally small size, high polarising power, high hydration energy and non availability of d-orbitals.

**Table 5.5 Comparison of properties of lithium with other elements of the group:**

| Lithium | Other elements of the family |
|---|---|
| Hard, high melting and boiling point | Soft and Lower melting and boiling point |
| Least reactive (For example it reacts with oxygen to form normal oxide, forms peroxides with great difficulty and its higher oxides are unstable) | More reactive |
| Reacts with nitrogen to give $Li_3N$ | No reaction |
| Reacts with bromine slowly | React violently |
| Reacts directly with carbon to form ionic carbides. For example 2Li + 2C --> $Li_2C_2$ | Do not react with carbon directly, but can react with carbon compounds. Na + $C_2H_2$ --> $Na_2C_2$ |
| Compounds are sparingly soluble in water | highly soluble in water. |
| Lithium nitrate decomposes to give an oxide | decompose to give nitrites |

**Table 5.6 Similarities between lithium and Magnesium**

| S.No. | Properties |
|---|---|
| 1 | Both lithium and magnesium are harder than other elements in the respective groups |
| 2 | Lithium and magnesium react slowly with water. Their oxides and hydroxides are much less soluble and their hydroxides decompose on heating. |
| 3 | Both form a nitride, $Li_3N$ and $Mg_3N_2$, by direct combination with nitrogen |
| 4 | They do not give any superoxides and form only oxides, $Li_2O$ and MgO |
| 5 | The carbonates of lithium and magnesium decompose upon heating to form their respective oxides and $CO_2$. |
| 6 | Lithium and magnesium do not form bicarbonates. |
| 7 | Both LiCl and $MgCl_2$ are soluble in ethanol and are deliquescent. They crystallise from aqueous solution as hydrates, $LiCl{\cdot}2H_2O$ and $MgCl_2{\cdot}8H_2O$ |

### Diagonal Relationship:

Similarity between the first member of group 1 (Li) and the diagonally placed second element of group 2 (Mg) is called diagonal relationship. It is due to similar size (r $Li^+$ = 0.766 Å and $Mg^{2+}$ = 0.72 Å) and comparable electronegativity values (Li = 1.0; Mg = 1.2).